

Rithwik Gupta

MTS @ Etched | EE + CS @ Stanford | rithwikca2020@gmail.com | 408-442-8704 | rithwikg.com | [Github](https://github.com) | [Linkedln](https://www.linkedin.com/in/rithwikg)

EXPERIENCE

Member of Technical Staff, Architecture & Inference | [Etched](#)

June 2025 - Present

- Worked full-time over the summer and part-time during school (currently full-time)
- Trained and implemented **speculative decoding models (DFlash, EAGLE)** and **frontier models (Deepseek, Kimi K3)**; wrote multichip kernels, quantized, and post-trained speculators to optimize performance on Etched hardware (**3-4x speedups**)
- Heavily optimized **end-to-end inference with advanced speculation** (both kernels and host overhead)
- Wrote kernels for various parts of transformer functionality, namely per-head norms, GQA, and RoPE
- Implemented early scheduling and memory (KV Cache) management
- Wrote Rust, C++, and Python

USA Computing Olympiad (USACO) Finalist and 2x USA Top 10

January 2021 - December 2025

- Placed **6th** nationally out of 10,000 participants in a USACO contest
- One of 25 students invited to attend [USACO Camp](#) to compete for the US [IOI](#) Team (June 2024)
- Developed thorough understanding of how to apply advanced algorithms and data structures

LLM Robustness Research | UC Santa Cruz [Information Retrieval Lab](#)

July 2023 - December 2025

- Built a dataset to address the robustness of RAG systems when provided with false documents
- Published at **NeurIPS 2025 main conference (25 citations)**, presented at the ACL FEVER workshop and Baylearn
- Started by researching retrieval augmented political debate summarization at the summer program [COSMOS](#)

Applied AI Research | Under [Dr. Daniel Muthukrishna at MIT](#)

June 2023 - Present

- Published 2 research papers ([One](#), [Two](#), [Three](#), [pending](#), **35 citations**) related to the field of AI applied to astronomy; presented at 4 ICML Workshops ([AI4Science](#), [FMSD](#), [ML4Astro1](#), [ML4Astro2](#))
- Built novel anomaly detection pipeline (currently running on real telescopes); ICML (International Conference on Machine Learning) presentation, RAS publication
- Worked on transfer learning for classification and built domain-agnostic foundation models
- Models currently running on [real telescopes](#)

SPARC (Summer Program on Applied Rationality and Cognition) (< 2.5% Acceptance)

July 2025

5x AIME Qualifier (American Invitational Mathematics Exam)

2021 - 2026

COSMOS AI Cluster (<4% Acceptance)

July 2023

PUBLICATIONS

Gupta, R. et al. Worse than Zero-shot?... 2025, **Neural Information Processing Systems (NeurIPS)**. [[ArXiv](#)]

Gupta, R. et al. Building Foundation Models... 2025, in review at MNRAS. [[ICML](#)]

Gupta, R. et al. Transfer Learning for Transient Classification.... 2025, MNRAS. [[ArXiv](#)]

Gupta, R. et al. A Classifier-Based Approach to Multi-Class Anomaly Detection... 2025, RASTI. [[ArXiv](#)]

NOTABLE PROJECTS

[Rubik's Cube Solver](#): Robot that physically solves Rubik's Cubes (currently building v2 for 4x4s)

[astromcad](#): Astronomical anomaly detection Python package with 50,000+ downloads (implements my research)

[Games](#): Collection of common games and AI algorithms to play them (e.g. Snake, Connect 4)

[Competitive Programming Bot](#): LLM-based application to provide hints for competitive programming problems

Lightning: RAG-based application which helps students take more effective notes by grounding them to a syllabus

'varhist' Python Package: Developed a debugging package with 20,000+ downloads

EDUCATION

EE + CS @ Stanford

LEADERSHIP

Founder and President (Irvington Competitive Programming Club)

November 2022 - June 2026

President (Irvington Math Club)

June 2023 - June 2026

Founder and Team Captain (FTC Team #18504; 2x Regional Qualifiers)

August 2022 - Dec 2024